Photography Basics

Knowledge document 087 of 100

Introduction to Photography

Photography is the art, science and practice of creating images by recording light, either electronically with an image sensor or chemically on film. The word comes from the Greek words "photos", meaning light, and "graphein", meaning to draw, so photography means drawing with light. A camera works like the human eye: a lens focuses light from a scene onto a light-sensitive surface, and the amount and quality of light that reaches the surface decides how the picture looks. Today most photographs are taken with smartphones, and photography is important in journalism, science, medicine, advertising, travel and social media.

Brief History

The first permanent photograph was made by the French inventor Nicephore Niepce around 1826 to 1827. In 1839, Louis Daguerre announced the daguerreotype process, and 19 August is celebrated as World Photography Day. William Henry Fox Talbot invented the negative-positive process, which allowed many prints. George Eastman introduced the Kodak camera in 1888 with the slogan "You press the button, we do the rest". In 1975 Steven Sasson, an engineer at Kodak, made the first digital camera. Smartphone cameras became common after about 2007. In India, early photographers include Lala Deen Dayal, in the 19th century, and Homai Vyarawalla, the first Indian woman photojournalist, while Raghu Rai and Raghubir Singh are renowned. The Madras Photographic Society was formed in 1856, and the British photographer Linnaeus Tripe photographed monuments of the Madras Presidency in the 1850s.

The Exposure Triangle

Exposure is the amount of light that reaches the camera's sensor. It is controlled by three settings called the exposure triangle: aperture, shutter speed and ISO. A good exposure is neither too dark (underexposed) nor too bright (overexposed). Changing one setting requires adjusting the others to keep the exposure the same.

What Are Aperture, Shutter Speed and ISO?

Aperture is the size of the opening in the lens through which light passes, like the pupil of the eye. It is measured in f-numbers (f-stops), such as f/1.8, f/2.8, f/5.6, f/8 and f/16. A small f-number, such as f/1.8, means a large opening, letting in much light and creating a blurred background; a large f-number, such as f/16, means a small opening, letting in less light and giving more of the scene in focus. Aperture is the main control of depth of field.

Shutter speed is the length of time the camera's shutter stays open to expose the sensor to light, measured in seconds or fractions of a second, such as 1/1000,

1/250, 1/60 or 2 seconds. A fast shutter speed, such as 1/1000 second, freezes motion, as in sports and birds in flight, and lets in less light. A slow shutter speed, such as 1/30 second or longer, lets in more light and creates motion blur, as in silky waterfalls and light trails at night, but needs a tripod to avoid camera shake.

ISO is the sensitivity of the camera's sensor to light. A low ISO, such as 100, is less sensitive and gives clean, sharp images in bright light. A high ISO, such as 1600 or 6400, is more sensitive and allows shooting in dim light but adds grain or noise to the image. Photographers keep ISO as low as possible and raise it only when needed. Most cameras and phones also have automatic and manual modes, such as aperture priority (A or Av), shutter priority (S or Tv) and manual (M).

What Is the Rule of Thirds?

The rule of thirds is a composition guideline in which the image is imagined as divided by two equally spaced horizontal lines and two equally spaced vertical lines, making a grid of nine equal parts, like a tic-tac-toe board. The important elements, such as the main subject or the horizon, should be placed along these lines or at the four points where they intersect, instead of right in the centre. This makes pictures more balanced, dynamic and interesting, because the eye is naturally drawn to these points. For example, in a landscape the horizon is placed on the top or bottom third line, depending on whether the sky or the land is more interesting, and in a portrait the subject's eyes are placed near the upper third line. Most phones and cameras can display a grid to help. The rule is a guideline, not a law, and rules can be broken, such as in symmetrical shots with a subject at the centre. Other composition techniques include leading lines, which guide the eye into the picture; framing, using doors or arches; symmetry and patterns; filling the frame; negative space; and changing the viewpoint, such as shooting from a low angle.

What Is Depth of Field?

Depth of field (DoF) is the range of distance in a photograph, from near to far, that appears acceptably sharp. A shallow depth of field has only a small part in focus, with the foreground and background blurred, which is popular in portraits because it isolates the subject; the blurred background is called bokeh. A deep depth of field has most of the scene sharp from front to back, which is preferred for landscapes, architecture and group photos. Depth of field depends on three factors: aperture (a wider aperture, with a smaller f-number, gives a shallower depth of field), focal length (a longer lens, such as a telephoto, gives a shallower depth) and distance to the subject (being close gives a shallower depth). Larger sensors produce shallower depth of field, so phones often use software "portrait mode" to simulate the blur.

Other Basics

Focal length, measured in millimetres, decides the angle of view: wide-angle lenses (such as 24 mm) capture more of the scene, and telephoto lenses (such as 200 mm) bring distant subjects closer. White balance adjusts colours so that white looks white under different lighting. The RAW format saves all the sensor data and allows more editing, while JPEG is a compressed format. Light is the most important element: the "golden hour", the hour after sunrise and before sunset, gives warm, soft light, while harsh midday sun creates strong shadows. A simple guide for sunny conditions is the "Sunny 16" rule: at f/16, set the shutter speed close to the reciprocal of the ISO. Tips for beginners: hold the camera steady, clean the lens, focus carefully, shoot many pictures, and learn editing with apps like Lightroom or Snapseed.

Photography in India and Chennai

India has a vibrant photography culture. Chennai hosts the Chennai Photo Biennale, held since 2019, as well as exhibitions and photo walks. Marina Beach, Mylapore temples, Mahabalipuram and Pondicherry are among the favourite subjects. The ethics of photography include taking permission before photographing people, respecting privacy and sacred places, and avoiding altering images to mislead.

Challenges and Future

Challenges include AI-generated images that make it hard to know what is real, and copyright. Computational photography, which uses software to merge multiple exposures, is making smartphone cameras more powerful.

Quick Facts

- Aperture is the lens opening (f-stop), shutter speed is how long the sensor is exposed, and ISO is the sensor's sensitivity to light.
- The rule of thirds divides the frame into a 3 by 3 grid and places subjects on the lines or intersections.
- Depth of field is the range of distance that appears sharp; a wide aperture gives a shallow depth of field.
- The first permanent photograph was made by Nicephore Niepce around 1826.
- World Photography Day is on 19 August.